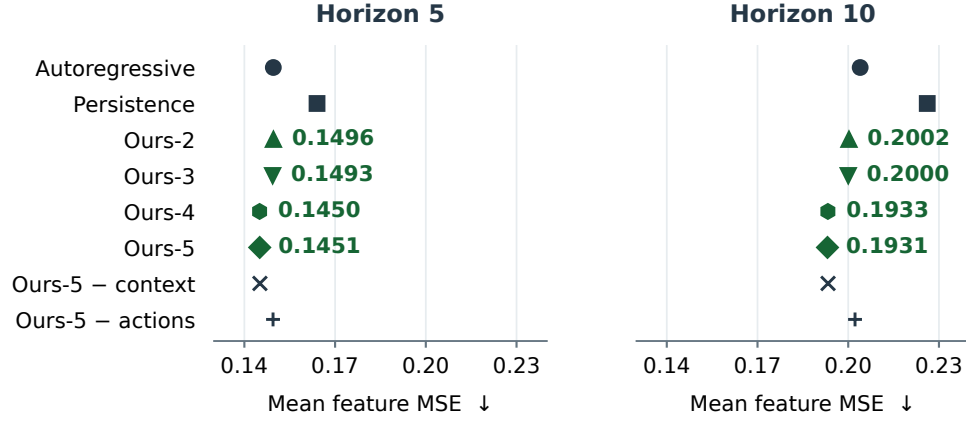
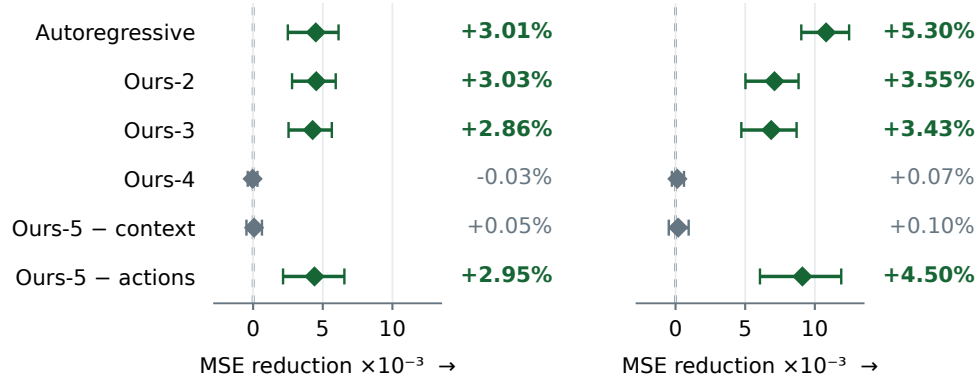

ID	Mechanism	Status
Ours-1	Original context-based ShiftWM	Completed; different protocol
Ours-2	Observed anchor + additive innovation	Completed
Ours-3	Observed anchor + bounded innovation	Completed; follow-up
Ours-4	Gated mixing + unbounded innovation	Completed; follow-up
Ours-5	Gated mixing + bounded innovation	Completed

Table 1: Stable mechanism identifiers, not a performance ranking. Ours-1 uses the original context-based protocol and is excluded from the spatial comparison axis. Context-off and action-free are Ours-5 ablations; autoregression and persistence retain baseline labels. Ours-3/4 completed their registered follow-up after the original controls were revealed; all six new models trained 30 epochs.

(a) Native spatial error



(b) Paired gains for Ours-5



Bars: 95% paired CI. Labels: relative point gains.

Figure 1: Completed spatial versions on original DROID validation. (a) Absolute native 4×4 standardized feature MSE at endpoints 5 and 10; green marks identify our mechanism variants, not a winner. (b) Ours-5 is the reference for paired reductions against six comparators; right of zero favors Ours-5. Only registered matching contrasts are used. Ours-4 and Ours-5 are close: both native comparison intervals include zero, and Ours-4 has the lower h5 point estimate. Bars show 95% paired recording-session $\times$ training-seed bootstrap intervals (10,000 draws; seed 173); percentages are relative point reductions. Gray intervals include zero; favorable green intervals exclude zero without multiplicity adjustment. All 21 spatial models completed 30 epochs: 15 original models plus 6 new component models; the four-arm factorial subset has 12 models including reused controls. Ours-3/4 were added in a registered follow-up after the original controls were revealed. Evaluation uses 1,631 matched windows, 141 episodes, 59 sessions and 3 seeds. Windows are averaged within episodes, then episodes and seeds equally. Ours-1 has a different protocol and remains off this axis. Companion tables preserve the original 16 comparisons and all 20 component effects, including four signed interactions and every unfavorable or inconclusive result. These are development comparisons, not independent-test or state-of-the-art claims.

Metric	Comparator	h	Relative gain	MSE reduction $\times 10^3$ [95% CI]
Native	Autoregressive	5	+3.008%	+4.499 [+2.491, +6.132]
Native	Autoregressive	10	+5.300%	+10.809 [+9.028, +12.461]
Native	Ours-2	5	+3.029%	+4.532 [+2.795, +5.930]
Native	Ours-2	10	+3.547%	+7.102 [+5.019, +8.831]
Native	Ours-5 – context	5	+0.048%	+0.069 [-0.478, +0.633]
Native	Ours-5 – context	10	+0.097%	+0.188 [-0.485, +0.943]
Native	Ours-5 – actions	5	+2.948%	+4.406 [+2.149, +6.551]
Native	Ours-5 – actions	10	+4.503%	+9.107 [+6.061, +11.896]
Original 2×2	Autoregressive	5	+0.441%	+0.656 [-1.918, +2.499]
Original 2×2	Autoregressive	10	+2.538%	+5.150 [+2.727, +7.196]
Original 2×2	Ours-2	5	+0.527%	+0.784 [-1.370, +2.281]
Original 2×2	Ours-2	10	+0.472%	+0.938 [-1.784, +2.897]
Original 2×2	Ours-5 – context	5	-0.194%	-0.286 [-1.016, +0.386]
Original 2×2	Ours-5 – context	10	-0.108%	-0.213 [-1.075, +0.644]
Original 2×2	Ours-5 – actions	5	+2.948%	+4.493 [+1.778, +7.086]
Original 2×2	Ours-5 – actions	10	+4.085%	+8.422 [+4.691, +11.892]

Table 2: All 16 completed spatial contrasts: Ours-5 versus each comparator. Positive reductions favor Ours-5; bold green percentages indicate positive point estimates, not statistical significance. Paired 95% recording-session and seed bootstrap intervals use 10,000 draws (seed173), without multiplicity correction. Both unfavorable coarse-coordinate context-ablation point estimates remain included. Native and original 2×2 coordinate errors are different metrics and cannot be pooled. Original validation only.

Comparison	Metric	h	Relative gain	Effect $\times 10^3$ [95% CI]
Ours-3 vs Ours-2	Native	5	+0.173%	+0.259 [-0.017, +0.535]
Ours-3 vs Ours-2	Native	10	+0.117%	+0.234 [-0.160, +0.649]
Ours-3 vs Ours-2	2×2	5	+0.185%	+0.275 [+0.009, +0.533]
Ours-3 vs Ours-2	2×2	10	+0.223%	+0.443 [+0.009, +0.907]
Ours-5 vs Ours-4	Native	5	-0.025%	-0.037 [-0.387, +0.309]
Ours-5 vs Ours-4	Native	10	+0.069%	+0.133 [-0.263, +0.606]
Ours-5 vs Ours-4	2×2	5	-0.074%	-0.109 [-0.559, +0.287]
Ours-5 vs Ours-4	2×2	10	-0.030%	-0.060 [-0.652, +0.449]
Ours-4 vs Ours-2	Native	5	+3.054%	+4.569 [+2.917, +5.870]
Ours-4 vs Ours-2	Native	10	+3.481%	+6.970 [+5.013, +8.666]
Ours-4 vs Ours-2	2×2	5	+0.601%	+0.893 [-1.162, +2.359]
Ours-4 vs Ours-2	2×2	10	+0.502%	+0.998 [-1.653, +2.923]
Ours-5 vs Ours-3	Native	5	+2.861%	+4.273 [+2.542, +5.650]
Ours-5 vs Ours-3	Native	10	+3.434%	+6.868 [+4.723, +8.690]
Ours-5 vs Ours-3	2×2	5	+0.343%	+0.509 [-1.618, +1.935]
Ours-5 vs Ours-3	2×2	10	+0.250%	+0.495 [-2.224, +2.513]
Interaction D	Native	5	—	+0.296 [-0.115, +0.778]
Interaction D	Native	10	—	+0.101 [-0.471, +0.710]
Interaction D	2×2	5	—	+0.384 [-0.061, +0.938]
Interaction D	2×2	10	—	+0.503 [-0.175, +1.344]

Table 3: All 20 predeclared component effects from the original-validation follow-up: six new models plus six reused controls, each trained for 30 epochs. The first four blocks report error reductions (second-model MSE minus first-model MSE): positive favors the first model. Bounding is isolated within the Ours-3/Ours-2 and Ours-5/Ours-4 pairs; mixing is compared within Ours-4/Ours-2 and Ours-5/Ours-3. The last block retains the separately signed interaction $D = (\text{MSE}_5 - \text{MSE}_4) - (\text{MSE}_3 - \text{MSE}_2)$: negative means bounding helps more with mixing; no percentage is defined for D . All four interaction intervals include zero. Bold green denotes positive pairwise point estimates, not significance. Paired session/seed 95% bootstrap intervals use 10,000 draws, seed 173, without multiplicity correction. This follow-up was designed after existing controls were revealed; mixing includes the gate, identity bias and approximately 1.84% extra active parameters.